

Root relationships-cubic, quartic

Key Points

For a cubic with roots α, β, γ

$$\Sigma\alpha = \alpha + \beta + \gamma = -\frac{b}{a},$$

$$\Sigma\alpha\beta = \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

For a quartic with roots $\alpha, \beta, \gamma, \delta$

$$\Sigma\alpha = -\frac{b}{a}, \quad \Sigma\alpha\beta = \frac{c}{a}$$

$$\Sigma\alpha\beta\gamma = -\frac{d}{a}, \quad \alpha\beta\gamma\delta = \frac{e}{a}$$

Basic Skills

Q1. For the quadratic $z^2 + 10z - 4 = 0$ with roots α, β

- a) Write down the value of $\alpha + \beta$ and $\alpha\beta$
- b) Find the related quadratic with roots $1 - \alpha, 1 - \beta$
- c) Find the related quadratic with roots $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

Q2. For the cubic $z^3 - z^2 + 10z - 4 = 0$ with roots α, β, γ

- a) Write down the value of $\Sigma\alpha$, $\Sigma\alpha\beta$ and $\alpha\beta\gamma$
- b) Find the value of $(2\alpha + 1) + (2\beta + 1) + (2\gamma + 1)$
- c) Find the value of $(2\alpha + 1)(2\beta + 1) + (2\alpha + 1)(2\gamma + 1) + (2\beta + 1)(2\gamma + 1)$
- d) Find the value of $(2\alpha + 1)(2\beta + 1)(2\gamma + 1)$

Competency Skills

Q3. For the cubic $z^3 - z^2 + 10z - 4 = 0$ with roots α, β, γ find the cubic with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$

Q4. For the cubic $z^3 - 5z + 4 = 0$ with roots α, β, γ

- a) Write down the value of $\Sigma\alpha$, $\Sigma\alpha\beta$ and $\alpha\beta\gamma$
- b) Hence find the cubic with roots $3\alpha - 1, 3\beta - 1, 3\gamma - 1$

Q5. Show that $(\Sigma\alpha)^2 = \Sigma\alpha^2 + 2\Sigma\alpha\beta$

Q6. For the cubic $z^3 - 5z^2 + 6z + 1 = 0$ with roots α, β, γ

- a) Hence find the cubic with roots $\alpha\beta, \alpha\gamma, \beta\gamma$
- b) Find the cubic with roots $\alpha^2, \beta^2, \gamma^2$

Advanced Skills

Q6. Given that a cubic has roots α, β, γ can you find the cubic with roots $\alpha^2, \beta^2, \gamma^2$ by substitution of z for z^2 . State your answer with reasoning.

Q7. For the quartic $z^4 - 3z^3 + 2z^2 - z + 2 = 0$ find the quartic equation with roots $2 - \alpha, 2 - \beta, 2 - \gamma, 2 - \delta$

Q8. The roots of the equation $2z^3 - 12z^2 + cz - 15 = 0$ are in geometric progression, solve the equation and find c

Q9. The quartic $z^4 + bz^3 + 30z + e = 0$ has roots $\alpha, 5\alpha, \beta, -\beta$. Find b and e

Q10. Find a condition for which two roots of a cubic equation are reciprocals of each other

Extension

Q11. Write down the root relations for a polynomial of degree n

Hint: Spot the pattern for the sums of different levels of multiplied roots

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